SUMMARY · MK 6 BUSINESS ANALYTICS · WEEK 1

Introduction to Business Analytics

Business analytics uses data, statistics and models for better decisions. Four levels of analytics, DIKW, the 14–session plan, and analytics supporting rather than replacing judgement. Case: your syndicate's PT BetonPrima West Java turnaround.

Material

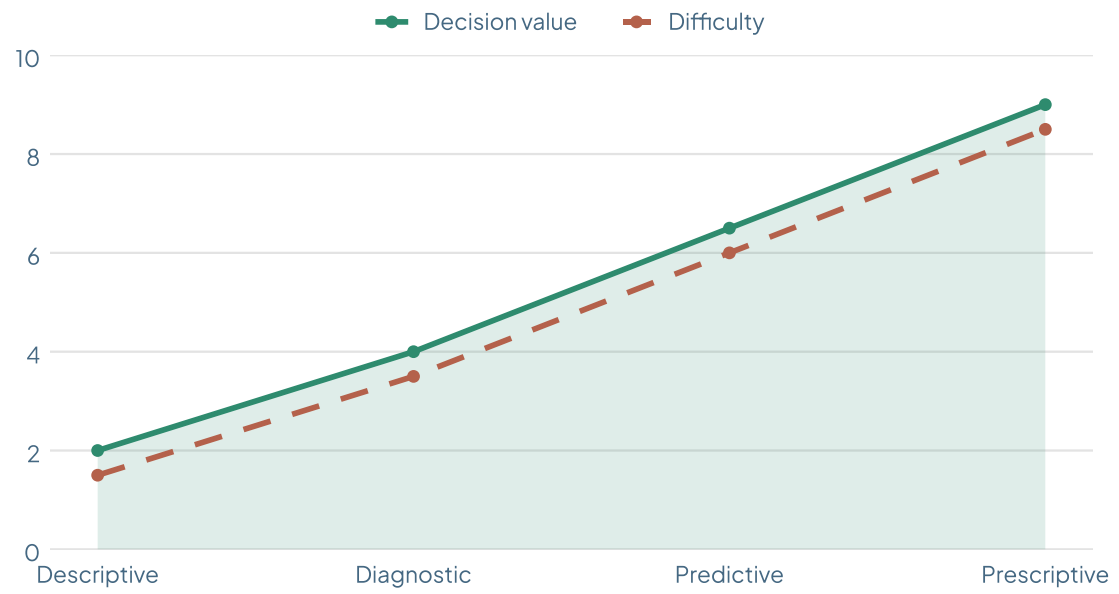
1 What business analytics is

Business analytics turns data into insight and action. Levels: descriptive, diagnostic, predictive, prescriptive — increasing difficulty and value. DIKW: Wamena temperatures → information → knowledge → wisdom (hold the festival in August).

2 Four levels of analytics: harder and more valuable

Deep dive

Value vs difficulty by level (conceptual)





3 Working principles

Start from the decision; expect messy data and conflicting definitions; analytics informs judgement; correlation is not causation; report ranges.

4 The analytics project workflow (CRISP-DM) and how to frame the question

The "start from the decision" principle is easiest to follow with a standard workflow. The most widely used in industry is **CRISP-DM** (Cross-Industry Standard Process for Data Mining):

1. Business understanding Which decision must be made, by whom, when, and how will success be measured?

2. Data understanding What data exists, where from, how complete, and how is it defined? (Week 2)

3. Data preparation Cleaning, joining and reshaping data. Often takes 60–80% of project time.

4. Modelling Descriptive, predictive or optimisation — matching the question. (Weeks 3–5)

5. Evaluation Does the result answer the business question, not just score well statistically?

6. Deployment The result is used in a real process: a dashboard, an approval rule, a production schedule. Then monitored and repeated.

The flow is not a straight line: teams often go back from step 2 to 1 (the data cannot answer the question) or from 5 to 3 (the model looks good but the data is biased).

A template for framing an analytics question:

Element	Question	BetonPrima example
Decision	What will be decided?	Choose between a price discount, 2 new depots or a new kiln
Decision owner	Who decides, and when?	The board, at next quarter's budget meeting
Success measure	Which number should change?	West Java volume back to 1.0m tonnes/quarter; on-time delivery ≥ 90%

Hypothesis	What is the suspected cause?	Customers leave because deliveries are late when it rains, not because of price
Data needed	Which data can test the hypothesis?	Delivery time per order + rainfall data + closed accounts + interviews
Action	What will we do differently if the hypothesis is true/false?	True → depots; false (really price) → consider targeted discounts

Signs an analytics project will fail

"We have lots of data, let's look for insights." Without a clear decision, analysis produces many interesting charts but nothing changes. If you cannot answer "what will we do differently after this?", the question is not ready yet.

5 BetonPrima in charts: four levels in one case

Deep dive

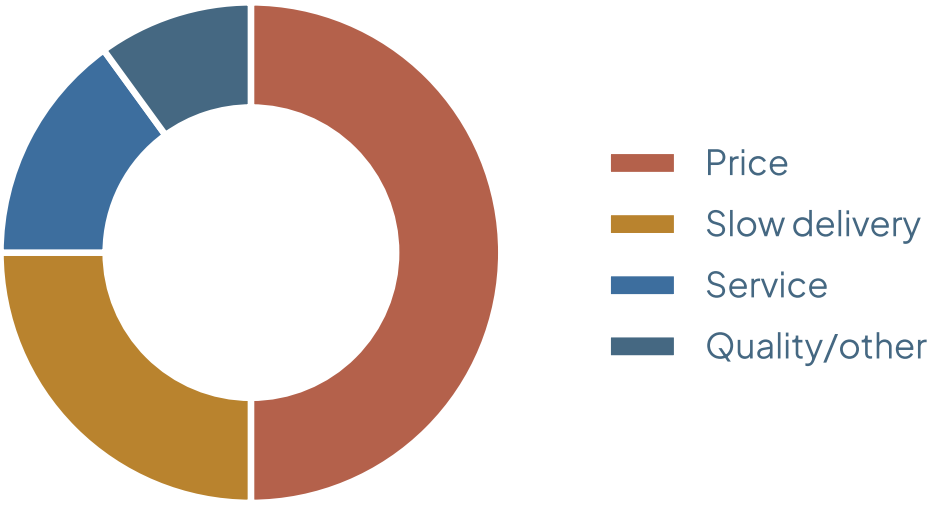
−20%
West Java volume

214
Accounts lost

5.4 days
Order-to-site

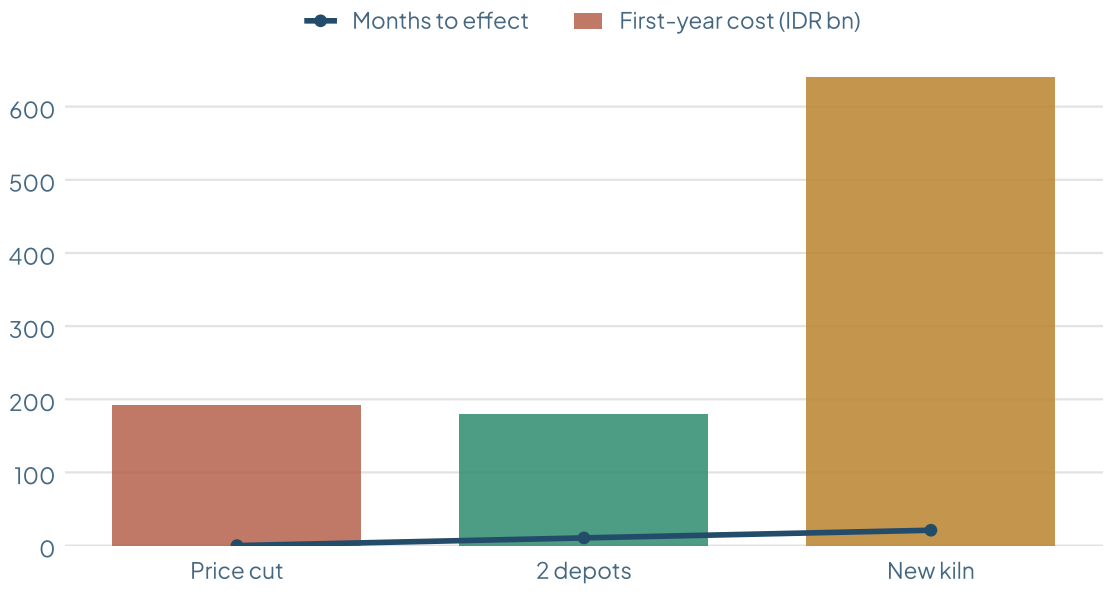
91% → 68%
On-time

Churn reasons in the system log



Interviews showed many 'price' entries were really 'late when raining'.

Three options: first-year cost and months to effect



Quick maths

Discounting at 48bn/quarter ≈ 192bn a year, more than two depots (180bn).

Key takeaways

1. Four levels.
2. DIKW.
3. Start from the decision.
4. Analytics supports judgement.
5. Mislabelled data hides root causes.

Case studies

Case & worked answers

Your syndicate: PT BetonPrima West Java turnaround

Descriptive: West Java volume fell 20% while demand was flat; 214 accounts closed. Diagnostic: churn logged as "price" was really rain-related late delivery (rigid drop-down — a capture problem); lead time 5.4 days vs 3.0, on-time 91% → 68%; weather and sales neglect ruled out. Prescriptive: two depots (IDR 180bn) beat price cuts and a kiln rebuild. Add: fix the capture form and compute depot payback.

Worked examples

Problem 1 **Easy** **Quick prescriptive maths: discount vs depots**

BetonPrima: the price-discount option costs IDR 48bn per quarter (recurring). The two-depot option costs IDR 180bn once. After how many months are the depots cheaper than the discount? What does this calculation miss?

1. Discount cost per month = 48 ÷ 3 = IDR 16bn.

2. ```
   Break-even = One-off cost ÷ Recurring cost per month = 180 ÷ 16 ≈ 11.25 months
   ```

3. After about 11 months the depots are cheaper — and they attack the root cause (rainy-day delivery) while the discount treats a symptom. Missed: the 9–12–month build time (customers may keep leaving), execution risk, and the time value of money.

Answer: About 11.25 months.

Practice questions

Try answering first. The answer key is below each question.

1. 'What should we do?' is…

A. Descriptive
B. Diagnostic
C. Predictive
D. Prescriptive

Answer: D. Output is a decision.
   ```

2. BetonPrima's misclassified reasons are a defect at…

A. Pipeline
B. Capture (the form)
C. Dashboard
D. Model

Answer: B. A rigid single-choice drop-down.

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3. The option that attacks the root cause…

A. Price cut
B. Two depots
C. New kiln
D. More sales staff

Answer: B. Root = logistics bottleneck.

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4. 'August in Wamena tends to be warm' is…

A. Data
B. Information
C. Knowledge
D. Wisdom

Answer: C. A general pattern.

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5. Analytics should start from…

A. Available data
B. The decision to be made
C. The newest tool
D. A dashboard

Answer: B. What, when, what changes.

Sources

- WEEK 1 MK06 P3MD (slide berupa gambar)
- Course plan MK6
- PT BetonPrima Analytics — Syndicate 3 Kelas E (tugas kelompokmu)